

Tarun Tyagi

812-391-6150 | taruntyagi0193@gmail.com | linkedin.com/in/tarun-tyagi-018a4549

PROFESSIONAL SUMMARY

Full Stack Software Engineer with 8+ years of experience building scalable, cloud-native applications and high-performance web platforms. Expert in ReactJs, JavaScriptTypeScript, Node.js, and modern API architectures, with hands-on experience developing frontend applications, backend middleware services, and REST/GraphQL APIs. Skilled in microservices design, database optimization (DynamoDB, MongoDB, SQL), and cloud infrastructure on AWS. Specialized in designing automated evaluation pipelines for LLM and AI Agents using AWS Bedrock and promptfoo, fully integrated into CI/CD workflows. Proven track record delivering high-availability enterprise systems with 99.9% uptime serving 50k+ daily users, with measurable improvements in system performance, reliability, and deployment automation.

TECHNICAL SKILLS

Languages: JavaScript (ES2022+), TypeScript, NodeJs, Python, SQL

Frontend: React.js, Redux, Vue.js, D3.js, HTML5, CSS3

Backend: REST APIs, GraphQL, Microservices, AWS Lambda

AI / LLM: AWS Bedrock Agents, Prompt Engineering, promptfoo (LLM Regression Testing), Claude API, CoPilot

Cloud & DevOps: AWS (EC2, Lambda, S3, API Gateway, CloudWatch, Elasticsearch, Route 53), Terraform, CI/CD Pipelines, GitOps

Databases: DynamoDB, MongoDB, MSSQL, Redshift

Testing & Observability: ViTest, Jest, Pytest, JMeter, Dynatrace, AWS CloudWatch

WORK EXPERIENCE

State Farm (ARV Systems) | Bloomington, IL

October 2020 – Present

Software Engineer — Full Stack & AI Platform

- Designed and developed scalable full stack applications using React.js, Redux, TypeScript, Node.js, and REST APIs, supporting enterprise platforms with 50k+ daily users and 99.9% uptime SLA.
- Built 25+ serverless microservices on AWS Lambda (Node.js ES2024+) powering real-time, historical, and operational dashboard APIs with a layered Handler to Business Object to DAO architecture and integration with DynamoDB, Redshift, and S3.
- Designed and implemented automated evaluation pipelines for AWS Bedrock AI Agents integrating regression testing into CI/CD using promptfoo and LLM as judge scoring, reducing AI regression escapes to production by 60%.
- Engineered multi-region active-active backend architecture using Route 53 failover across US East and US West, improving availability and disaster recovery.
- Implemented JWT-based custom authorizer in Node.js for AWS API Gateway, enabling secure centralized authentication across all backend services.
- Built responsive UI screens and a reusable React component library, optimizing frontend rendering and reducing page load times by 30%.
- Authored Infrastructure as Code using Terraform for consistent, repeatable multi-environment deployments across dev, staging, and production.
- Established Git-based CI/CD pipelines covering automated builds, unit testing, security scanning, artifact versioning, and zero-downtime deployments.
- Mentored multiple junior developers, led onboarding, conducted regular code reviews, and provided hands-on guidance across Terraform, TypeScript, React.js, Node.js, and AWS, accelerating their ramp-up and raising overall team code quality.

- Monitored system performance and application health using Dynatrace and AWS CloudWatch; designed logging, alerting, and failover strategies to support proactive incident response.
- Executed comprehensive unit testing using ViTest and conducted rigorous load and performance testing with JMeter to identify architectural bottlenecks and validate system reliability.

Blinkpad Inc. | San Jose, CA

July 2020 – October 2020

Web Development Engineer

- Developed a full stack job marketplace platform for healthcare and hospitality industries using Vue.js, TypeScript, and Python, implementing role-based access control to serve two distinct user personas (Business and Candidate) with separate routing, views, and API scopes.
- Built dynamic Business and Candidate profile modules with multi-section forms, real-time field validation, and structured API integrations supporting job posting workflows, professional history management, and industry-specific user data.
- Engineered multi-resume upload feature allowing candidates to store multiple resumes linked to their profile in MongoDB and select a specific resume at job application time, handling file associations and conditional payload construction on submission.
- Designed and integrated a media upload pipeline using AWS S3 with presigned URLs for secure direct uploads of video and audio clips, managing S3 object references within user profiles in MongoDB.
- Designed flexible MongoDB schemas for industry-specific Business and Candidate profiles; optimized queries using indexing and projection, improving data retrieval performance and reducing response times.
- Wrote Jest unit tests for Vue.js components and performed end-to-end testing and debugging, improving application stability and reducing production defects.
- Conducted API performance and stress testing using JMeter to ensure the application could scale efficiently during peak traffic periods.

Aon Hewitt | India

June 2016 – October 2017

Graduate Engineer — Frontend

- Built a client-facing React.js and Redux dashboard enabling enterprise clients like Walmart and Enterprise Holdings to visualize aggregated employee sentiment data, designing scalable multi-tenant component architecture with isolated data views.
- Implemented interactive data visualizations using D3.js including trend charts and word cloud components to surface key themes from large volumes of employee survey responses.
- Optimized React rendering and Redux state management to efficiently handle high-volume datasets, reducing unnecessary re-renders and improving dashboard responsiveness.
- Utilized Jest for comprehensive unit testing of frontend states and logic, alongside JMeter for simulation testing to benchmark page-load and backend response speeds.
- Engineered backend data processing pipelines to handle 100,000+ survey responses per client, optimizing complex SQL queries using joins, indexing, and aggregations for efficient data retrieval.
- Automated validation of large client survey datasets using Excel VBA macros, enforcing data integrity rules and eliminating hours of manual effort per data cycle.
- Built ETL pipelines using Microsoft SSIS to ingest validated data from Excel into structured reporting databases, ensuring reliable data flow for analytics and reporting.

Wipro | India

May 2015 – June 2016

Frontend Developer

- Developed responsive web applications using React.js, JavaScript, jQuery, and Bootstrap, implementing reusable component architecture to ensure cross-browser compatibility and consistent UI rendering across devices.
- Integrated frontend components with REST APIs using asynchronous data fetching patterns, handling loading states, error boundaries, and dynamic data rendering across dashboards and form-based workflows.

- Optimized DOM rendering and JavaScript execution by reducing reflows, minimizing redundant event listeners, and deferring non-critical scripts, improving page load performance and runtime responsiveness.
- Implemented client-side form validation, data sanitization, and structured error handling to enforce data accuracy and enhance user experience across input-heavy interfaces.
- Authored foundational Jest unit tests to validate core frontend logic and utilized JMeter to analyze component response latency under high user simulation.
- Collaborated in code reviews, maintained Git branching strategies, and authored technical documentation to support team collaboration and long-term maintainability of the codebase.

EDUCATION

Master of Science in Software Engineering

August 2018 – May 2020

San Jose State University, San Jose, CA | GPA: 3.73 / 4.00

Bachelor of Technology in Computer Science & Engineering

August 2011 – May 2015

MDU University, India | GPA: 3.90 / 4.00